

Chapter 3: Proteins

The Shape and Structure of Proteins

The Shape of a Protein Is Specified by Its Amino Acid Sequence

- Proteins are polymers of amino acids linked by covalent **peptide bonds** to form a **polypeptide**. The repeating sequence of atoms along the chain is the **polypeptide backbone** (Figure 3-1).
- The **amino acid sequence** is the unique order of the 20 common amino acids, each with a distinct **side chain** that confers specific chemical properties (Figure 3-2, Panel 3-1).
- Amino acids can be classified into four groups based on their side chains:
 - **Acidic**: Aspartic acid, Glutamic acid (negatively charged).
 - **Basic**: Arginine, Lysine, Histidine (positively charged).
 - **Uncharged polar**: Asparagine, Glutamine, Serine, Threonine, Tyrosine (hydrophilic).
 - **Nonpolar**: Alanine, Glycine, Valine, Leucine, Isoleucine, Proline, Phenylalanine, Methionine, Tryptophan, Cysteine (hydrophobic).
- The polypeptide chain has directionality, with an **amino terminus (N-terminus)** and a **carboxyl terminus (C-terminus)**.

Proteins Fold into a Conformation of Lowest Energy

- The 3D conformation of a protein is determined by its amino acid sequence and is stabilized by a large number of weak **noncovalent bonds** (Figure 3-4):
 - **Hydrogen bonds**: Between backbone atoms, side chains, and backbone-to-side-chain atoms.
 - **Electrostatic attractions**: Between charged side chains.
 - **Van der Waals attractions**: Between any two atoms at close range.
- A critical driving force is the **hydrophobic clustering force**, where nonpolar side chains are forced together in the protein's interior to minimize their disruptive effect on the hydrogen-bonded network of water molecules (Figure 3-5).
- The final folded structure, or **conformation**, is the one that minimizes the protein's free energy (

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- **Denaturation** (unfolding) and **renaturation** (refolding) experiments show that the primary sequence contains all the information for folding.
- **Molecular chaperones** assist in folding by preventing aggregation and guiding proteins along the most energetically favorable pathway.
- The flexibility of the polypeptide chain is constrained by **steric limitations**, restricting the possible phi (φ) and psi (Ψ) bond angles, as visualized in a **Ramachandran plot** (Figure 3-3).
- A folded protein is not static but exists as an ensemble of closely related substructures, or **conformers** (Figure 3-12, illustrated with ubiquitin).

Common Folding Patterns: The α Helix and the β Sheet

- These regular folding patterns, known as **secondary structure**, arise from hydrogen bonds between N-H and C=O groups in the polypeptide backbone (Figure 3-6).
- **α -helix**: A rigid, right-handed cylinder with a hydrogen bond between every fourth amino acid. It has a complete turn every 3.6 amino acids (Figure 3-23).
- **β -sheet**: A rigid structure formed when segments of the polypeptide chain lie side by side, connected by hydrogen bonds. Strands can be **parallel** or **antiparallel** (Figure 3-7).
- **Coiled-coil**: A stable structure formed when two or more α -helices wrap around each other, often with their nonpolar side chains interlocked (Figure 3-8).

Levels of Protein Organization

- **Primary structure**: The amino acid sequence.
- **Secondary structure**: α -helices and β -sheets.
- **Tertiary structure**: The full 3D conformation of a single polypeptide chain.
- **Quaternary structure**: The arrangement of multiple polypeptide chains (**subunits**) in a protein complex.
- Protein structure can be visualized in several ways (Figure 3-9):
 - **Backbone model**: Shows the overall organization of the polypeptide chain.
 - **Ribbon model**: Emphasizes secondary structures like α -helices and β -sheets.
 - **Wire model**: Shows all atoms and side chains, useful for predicting ligand binding.

- **Space-filling model:** Shows the protein's surface and contour.

Protein Domains and Families

- A **protein domain** is a segment of a polypeptide (typically 40-350 amino acids) that folds independently into a stable structure (Figure 3-10). Large proteins are often modular, composed of multiple domains linked by intrinsically disordered sequences (Figure 3-11).
- Common domains like the SH2 domain or the homeodomain have conserved structures (Figure 3-16).
- **Protein families** consist of proteins with similar sequences and structures, derived from a common ancestor (e.g., serine proteases, Figure 3-13; homeodomains, Figure 3-14).
- Evolution creates new proteins through **domain shuffling**, combining existing domains in new arrangements (Figure 3-15). This can lead to extended structures, like in fibronectin (Figure 3-17), or proteins with related functions (Figure 3-18).

Protein Assemblies

- Proteins assemble into larger structures using noncovalent bonds.
- Complexes can be formed from identical subunits (**dimers**, **tetramers**) or different subunits (**heteromers**). Assemblies can form rings, helices, or more complex shapes (Figure 3-19, 3-21, 3-26).
- **Hemoglobin** is a tetramer of two α -globin and two β -globin subunits (Figure 3-20).
- Globular proteins like actin can polymerize into long **filaments** (Figure 3-22).
- **Fibrous proteins** like collagen form long, triple-helical fibers that provide structural support (Figure 3-24).
- **Disulfide bonds** (S-S bonds) are covalent cross-links between cysteine residues that reinforce the structure of many extracellular proteins (Figure 3-25).
- **Viral capsids** are large protein assemblies that protect the viral genome, forming spherical (icosahedral) or helical structures (Figure 3-27, 3-28, 3-29).
- Some proteins, like insulin, are synthesized as inactive precursors and activated by **proteolytic cleavage** (Figure 3-30).

Protein Misfolding and Disease

- **Amyloid fibrils** are insoluble, self-propagating β -sheet aggregates associated with diseases like Alzheimer's and Parkinson's (Figure 3-31, 3-32).

- **Prion diseases** are caused by an infectious, misfolded protein (PrP^{*}) that converts normal PrP protein to the pathological form (Figure 3-33).
- Amyloid structures can also have normal functions, such as storing hormones (Figure 3-34).

Protein Function

How Proteins Work

- A protein's function depends on its ability to bind specifically to other molecules (**ligands**) at its **binding site** (Figure 3-35, 3-36).
- Binding specificity is achieved through the formation of many weak noncovalent bonds and precise molecular complementarity (Figure 3-41).
- The strength of binding is related to the equilibrium constant (K) and the standard free-energy change (ΔG°) (Figure 3-42). Small changes in the number of weak bonds can drastically alter binding affinity (Figure 3-43).
- The **evolutionary trace method** can identify functionally important sites on a protein's surface by mapping conserved residues (Figure 3-38).
- Proteins can bind to each other in various ways: surface-string, helix-helix, or surface-surface interactions (Figure 3-39).
- **Antibodies** are proteins with highly specific antigen-binding sites, demonstrating the principles of molecular recognition (Figure 3-40).

Enzymes as Catalysts

- **Enzymes** are proteins that catalyze reactions by lowering the **activation energy** (Figure 3-45). They can accelerate reactions by factors of a million or more (Figure 3-46).
- The **active site** of an enzyme binds the **substrate** and contains reactive amino acids that facilitate catalysis (Figure 3-37).
- Catalytic strategies include orienting substrates, straining bonds, and participating directly in the reaction (acid-base catalysis) (Figure 3-47, 3-50).
- **Lysozyme** is an example that cleaves a polysaccharide chain by distorting the substrate into its transition state (Figure 3-48, 3-49).

- Many enzymes use nonprotein **coenzymes** like retinal or heme to perform specific functions (Figure 3-51).
- **Enzyme kinetics** are described by **V_{max}** (maximum velocity) and **K_m** (Michaelis constant), which reflects substrate binding affinity (Figure 3-44).

How Proteins Are Controlled

- **Allosteric regulation:** The binding of a regulatory molecule to a site other than the active site causes a conformational change that alters the protein's activity.
 - **Feedback inhibition** is a common form of negative regulation (Figure 3-52, 3-53).
 - Regulation can be positive (activation) or negative (inhibition) through **conformational coupling** between binding sites (Figure 3-54, 3-55).
 - **Cooperative allosteric transitions** in multisubunit enzymes lead to a sharp, switch-like response to ligand concentration (Figure 3-56, 3-57).
- **Protein phosphorylation:** A major regulatory mechanism where a **protein kinase** adds a phosphate group and a **protein phosphatase** removes it. This acts as a molecular switch (Figure 3-58). The human genome contains hundreds of protein kinases (Figure 3-59).
- The **Src kinase** is an example of a protein regulated by phosphorylation at multiple sites, acting as a signal integrator (Figure 3-60, 3-61, 3-62).
- **GTP-binding proteins** (like Ras) act as molecular switches, cycling between an active (GTP-bound) and inactive (GDP-bound) state (Figure 3-63, 3-64).
- **Covalent modification by ubiquitin:** The attachment of ubiquitin marks proteins for different fates. Polyubiquitylation often targets proteins for degradation (Figure 3-65). The process involves E1, E2, and E3 enzymes (Figure 3-66, 3-67).
- **Motor proteins** and **protein machines** use the energy from ATP or GTP hydrolysis to drive ordered conformational changes, producing directional movement or carrying out complex tasks (Figure 3-68, 3-69, 3-70, 3-71, 3-72).
- **Multisite modification** allows for complex, combinatorial control of protein function (Figure 3-81).

Biomolecular Condensates

- Large, dynamic assemblies of proteins and RNA that form liquid-like droplets through **liquid-liquid phase separation** (Figure 3-80).
- They are held together by weak, multivalent interactions, often involving **intrinsically disordered protein (IDP)** regions (Figure 3-73, 3-77).
- IDPs can also act as flexible tethers or form elastic structures (Figure 3-74, 3-75).
- Condensates, such as the nucleolus or DNA repair factories, concentrate specific molecules to enhance reaction rates or create specialized environments (Figure 3-76, 3-78, 3-79).

Designing Proteins

- Understanding protein structure and function allows for the design of novel proteins with new functions, such as biosensors or catalysts (Figure 3-83).
- Large-scale analysis of protein interactions reveals complex cellular networks (Figure 3-82).